MentisDB

Durable, Verifiable Memory for Software Agents

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master after 0.10.10.55 (unreleased changes included) — 2026-11-01

Abstract

Agents forget. A coding agent that learned last month that the user wants tabs, that the build breaks on a certain linker flag, or that a design was rejected, starts the next session without any of it unless something durable and searchable carries it forward. MentisDB is that memory, built as an embedded Rust library and an optional single-binary daemon. Memory is an append-only log of typed records called *thoughts*. Each record is linked to its predecessor by a SHA-256 hash, can carry an Ed25519 signature bound to its chain and position, and can be governed by an access policy recorded inside the chain itself. A policy can make parts of a chain private: writable only with an authorized key, and readable only by a client that proves it holds one.

Retrieval combines BM25 lexical scoring with document-frequency gating, dense-vector similarity from a local ONNX model, typed graph expansion, append-order session cohesion, and optional Reciprocal Rank Fusion. Nothing is deleted: a superseded thought is replaced in results by its current successor, and one that a newer thought merely contests is shown with the contester ranked above it. A *standing context* query returns what is durably true about the user and the project regardless of age, next to the usual window of recent thoughts. None of this needs an external database, a network service or an LLM call.

With its current defaults and a small local embedding model (MiniLM, 384 dimensions), the pipeline reaches 76.7% Recall@10 on LoCoMo-10P and 69.8% Recall@5 on LongMemEval, up from 73.5% and 68.0% in release 0.10.9.54 under the same protocol. On real project chains, an audit found no loss of recall with the age of a memory, but found that near-duplicate thoughts crowded results and that paraphrased queries recalled far less than verbatim ones. Length-independent vector fusion and result-time duplicate collapse were built in response, and they raised rank-1 recall on the most duplicated chain from 75% to 98%. This paper describes the design as it is implemented, states each default exactly, and lists what is still weak.

1. The Problem and the Requirements

An agent's context window is working memory. It is erased at the end of each session and compacted during long ones. What the agent needs on top of it is the equivalent of long-term memory: a store it writes to as it works and reads from when it starts or when a question touches the past. We set eight requirements for that store.

1. **Durable.** An acknowledged write survives a process crash. With an opt-in setting it also survives power loss.
2. **Recallable regardless of age.** A preference stated three months ago must reach the agent as reliably as one stated yesterday. A window of recent records does not satisfy this on its own.
3. **Searchable by meaning and by words.** Users paraphrase. Exact identifiers, file names and error strings also matter. Both kinds of query must work.
4. **Current.** When a fact changes, the new version must win and the old one must stop surfacing, without destroying the history that explains the change.

5. **Attributable and verifiable.** Every record names the agent that wrote it. Edits to history are detectable, and a chain can require that writes carry valid signatures.
6. **Local and dependency-free.** The core path runs on a laptop with no database server, no cloud API and no per-write LLM cost.
7. **Portable across agents.** Claude Code, Codex, Copilot, Gemini and any other MCP client share one memory.
8. **Inspectable.** Every ranking can be explained, and every memory can be read, exported and audited by a human.

Common approaches fail at least one requirement. A Markdown notes file is durable and portable but is not searchable at scale and is loaded whole or not at all. Provider session memory is not portable and not inspectable. A plain vector store answers paraphrases but has no notion of currency, attribution or integrity, and misses exact identifiers. Systems that extract memories with an LLM on every write add latency, cost and a second model whose mistakes become permanent.

MentisDB addresses the requirements with three ideas. Memory is an append-only, hash-chained log, so history is never rewritten and changes are recorded as new thoughts that point at old ones. Retrieval is a local, multi-signal ranking whose every component is exposed in the result. A written protocol, shipped as an agent skill, tells agents when to read, when to write and what to link.

2. Architecture

MentisDB is a Rust crate (`mentisdb`) and one binary of the same name. The crate can be embedded directly. The binary runs a daemon and also serves as the command-line client.

```
Agents (Claude Code, Codex, Copilot, Gemini, ...)      Humans
      |  MCP (HTTP or stdio)    |  REST       |  CLI      |  browser
      v                         v             v           v
+-------------------------------------------------------------------+
| mentisdb daemon (one process, binds 127.0.0.1 by default)     |
|   MCP 9471   REST 9472   HTTPS MCP 9473   HTTPS REST 9474     |
|   Dashboard 9475 (HTTPS)                                      |
|   auth: optional bearer tokens; per-request read proofs       |
+-------------------------------+-------------------------------+
                                |
+-------------------------------v-------------------------------+
| MentisDb core (library)                                       |
|   append path: validate -> access policy -> sign check ->     |
|                relations/dedup -> hash -> writer thread       |
|   read path:   candidates (invalidation, as_of, filters) ->   |
|                BM25 + vectors + graph + cohesion -> rank      |
|   standing context, recent context, context bundles           |
|   agent registry, entity types, skill registry, webhooks      |
+-------------------------------+-------------------------------+
                                |
+-------------------------------v-------------------------------+
| Files under MENTISDB_DIR (default ~/.cloudllm/mentisdb)       |
|   canonical:  <stem>.tcbin (the hash chain)                   |
|   sidecars:   agents, entity types, vectors (+WAL, HNSW),     |
|               implicit-edge overlay   (all rebuildable)       |
|   global:     chain registry, skills, webhooks, tokens, TLS   |
+-------------------------------------------------------------------+
```

The canonical log is the only source of truth. Every other file is either a small registry written atomically or a derived index that can be rebuilt from the log.

Ports are set in `src/server.rs` (`MentisDbServerConfig`) and can be changed or disabled with `MENTISDB_MCP_PORT`, `MENTISDB_REST_PORT`, `MENTISDB_HTTPS_MCP_PORT`, `MENTISDB_HTTPS_REST_PORT` and `MENTISDB_DASHBOARD_PORT` (0 disables a listener). A self-signed TLS certificate is generated on first start under `tls/`.

3. Data Model

3.1 Thoughts

A thought is one memory record. The `Thought` struct (`src/lib.rs`) has these fields, in declared order. The order matters because records are encoded positionally.

Field	Type	Set by	Meaning
`schema_version`	`u32`	chain	Record layout version (currently 3)
`id`	`Uuid`	chain	Stable identifier (random v4)
`index`	`u64`	chain	Position in the chain, from 0
`timestamp`	`DateTime<Utc>`	chain	Commit time
`session_id`	`Option<Uuid>`	caller	Groups thoughts of one working session
`agent_id`	`String`	caller	Author, resolved through the agent registry
`signing_key_id`	`Option<String>`	caller	Key that signed this thought
`thought_signature`	`Option<Vec<u8>>`	caller	Detached Ed25519 signature (Section 4.2)
`thought_type`	`ThoughtType`	caller	What kind of memory this is
`role`	`ThoughtRole`	caller	How the system should treat it
`content`	`String`	caller	The memory text
`confidence`	`Option<f32>`	caller	0 to 1, clamped
`importance`	`f32`	caller	0 to 1, clamped; default 0.5
`tags`	`Vec<String>`	caller	Free-form labels, normalized; also carries the scope
`concepts`	`Vec<String>`	caller	Semantic labels, normalized
`refs`	`Vec<u64>`	caller	Back-references by index
`relations`	`Vec<ThoughtRelation>`	caller	Typed edges (Section 3.3)
`entity_type`	`Option<String>`	caller	Optional domain label, e.g. `bug_report`
`source_episode`	`Option<String>`	caller	Optional link to an external source
`prev_hash`	`String`	chain	Hash of the previous thought, empty for the first
`hash`	`String`	chain	SHA-256 of this record (Section 4.1)

The caller supplies the semantic fields. The chain assigns identity, position, time and the hash fields, so a client cannot choose where in history its record lands. For an unsigned thought the chain also normalizes input: it adds a `References` relation for each `ref`, fills a missing `valid_at` on relations with the commit time, and may add a `Supersedes` edge when automatic deduplication is on (Section 6.9). A signed thought is stored exactly as signed, with none of these additions, so its signature can be checked again later.

3.2 Types and Roles

`ThoughtType` says what kind of memory a thought is. It has 32 variants. In declared order they are:

> PreferenceUpdate, UserTrait, RelationshipUpdate, Finding, Insight, FactLearned, PatternDetected, Hypothesis, Mistake, Correction, LessonLearned, AssumptionIn-

validated, Constraint, Plan, Subgoal, Decision, StrategyShift, Wonder, Question, Idea, Experiment, ActionTaken, TaskComplete, Checkpoint, StateSnapshot, Handoff, Summary, Surprise, Reframe, Goal, LLMExtracted, AccessPolicy.

`LLMExtracted` marks memories produced by the optional extraction feature. `AccessPolicy` is reserved for governance records and is refused on the normal append path.

`ThoughtRole` says how the system treats a thought: `Memory`, `WorkingMemory`, `Summary`, `Compression`, `Checkpoint`, `Handoff`, `Audit` or `Retrospective`. Type and role are independent. A context-compaction checkpoint is a `Summary` thought with role `Checkpoint`. A post-mortem lesson is a `LessonLearned` thought with role `Retrospective`.

Types drive behaviour, not only labels. Standing context (Section 6.10) selects thoughts by type.

3.3 Typed Relations

A relation is a directed, typed edge from the new thought to an earlier one, possibly in another chain:

```
struct ThoughtRelation {
    kind: ThoughtRelationKind,
    target_id: Uuid,
    chain_key: Option<String>,        // set for cross-chain edges
    valid_at: Option<DateTime<Utc>>,  // when the edge starts to hold
    invalid_at: Option<DateTime<Utc>>,// when it stops holding
}
```

There are twelve kinds: `References`, `Summarizes`, `Corrects`, `Invalidates`, `CausedBy`, `Supports`, `Contradicts`, `DerivedFrom`, `ContinuesFrom`, `BranchesFrom`, `RelatedTo` and `Supersedes`. Because edges always point backwards in time, the relation graph of a chain is acyclic by construction.

Three kinds change what retrieval shows: `Supersedes`, `Corrects` and `Invalidates`. A thought that is the target of one of them is *invalidated* (Section 3.6).

3.4 Scopes, Sessions and Entities

A thought belongs to one memory scope: `User` (the default), `Session` or `Agent`. The scope is stored as a tag (`scope:user`, `scope:session`, `scope:agent`), so it is covered by signatures and filters like any tag. User scope holds facts about the person and the project. Session and agent scopes hold material private to one session or one agent, and standing context skips them.

`session_id` groups the thoughts of one working session. `entity_type` lets a chain declare domain types such as `bug_report` or `customer`; the per-chain entity-type registry records them. Both are filters. Neither changes ranking.

3.5 Agents

Thoughts carry only an `agent_id`. Everything else about an agent lives in a per-chain registry (`AgentRecord`): display name, owner, description, aliases, status (`Active` or `Revoked`), public keys, first and last seen index and time, and a thought count. Each public key records its id, algorithm (Ed25519 is the only one), key bytes, and when it was added and revoked.

A public key must be a valid, non-weak Ed25519 curve point. Keys are registered and revoked, and agents disabled, only by the operator (Section 4.3), never through REST or MCP. Through the library API a key also needs a proof of possession: a signature by that key over a payload with domain `mentisdb/key-registration/v1` that names the chain, agent, key id, key and a timestamp

within ten minutes (`KeyRegistrationProof::verify` in `src/access.rs`). When the operator registers a key, the operator vouches for it instead.

The registry is a sidecar file, not part of the hash chain. Section 4.3 explains what that means for trust.

3.6 Versions, Invalidation and Point-in-Time Reads

Nothing is deleted. When a thought is superseded, corrected or invalidated, the new thought carries the edge and the old one stays in the log.

At open, the chain scans all relations and builds the set of invalidated thought ids together with each target's invalidating edges (`observe_invalidating_relations`). Every append updates the set. The default read paths hide invalidated thoughts: filter search, ranked search, context bundles and recent context. A caller sees them with `include_invalidated=true`, which is how an agent audits what changed.

Hiding alone is not enough, for two reasons. A successor often uses different words than the thought it replaces, so hiding the old version can leave the agent with nothing. And agents rarely type their links. On one real installation, none of 14 `Correction` and `AssumptionInvalidated` thoughts and none of about 90 `Decision` thoughts carried a `Supersedes`, `Corrects` or `Invalidates` relation; half the corrections carried no link and the rest only plain references. The invalidated set stayed empty and the old policy kept winning searches. The lineage index (`src/lineage.rs`) addresses both problems. It is derived from the stored chain at open time, so existing chains benefit without migration.

- **Superseded.** A thought targeted by a strong edge is `superseded`. When it would have ranked, ranked search puts its *live successor* in its place: the newest unsuperseded end of the chain of supersessions, followed up to 64 steps. The successor inherits the old thought's score and reports which thoughts it stands in for (`successor_of`), so it reaches the agent even when its wording does not match the query.
- **Contested.** A weak link from a thought that says it replaces something marks the target `contested`. That covers a `Correction` or `AssumptionInvalidated` that refers to any thought, a `Decision` or `StrategyShift` that refers to an earlier decision, constraint, preference, plan or strategy with a plain reference or none, and any `Contradicts` edge. A contested thought stays visible, names the contesting thought, and that thought is promoted directly above it. Weak links do not hide, because a plain reference is often background and hiding it would lose valid memory.
- **Annotated everywhere.** Every thought returned on every surface carries `version_status` (`current`, `contested` or `superseded`) and, when not current, its `successors` and `current_version`, including results fetched with `include_invalidated=true`.
- **Advice at write time.** When a new `Correction`, `AssumptionInvalidated` or `Decision` looks like a replacement but carries no strong relation, the append response returns `warnings` naming up to three likely targets, so the agent can add the typed link.

Lineage follows cross-chain relations too, so a thought on a branch that supersedes one on its parent redirects results in branch and federated search, subject to the reader's audience. Context bundles redirect superseded seeds the same way. `with_resolve_successors(false)` on a library query restores plain hiding.

A query with `as_of = T` sees the chain as it was at time `T`. It considers only thoughts committed at or before `T`, and hides a thought only if an invalidating edge was in force at `T`: the edge's source was committed by then, its `valid_at` is not after `T`, and its `invalid_at` is not at or before `T`.

Live reads apply the same rule with `T` set to now, so an invalidating edge whose `invalid_at` has

passed, or whose `valid_at` lies in the future, does not hide its target.

3.7 Chains, Branches and Merges

A chain is one independent log, named by a chain key. Keys are validated at every entry point: not blank, at most 256 bytes, no control characters (`validate_chain_key`). The chain registry (`mentisdb-registry.json`) lists chains with their schema version, storage adapter, thought and agent counts, and location.

`branch_from` creates a new chain whose first thought is a `StateSnapshot` with a cross-chain `BranchesFrom` edge to a thought of the source chain. Nothing is copied and the source is untouched; the branch inherits context by reference. Ranked search through the daemon follows `BranchesFrom` edges and also searches the ancestor chains, deduplicating by thought id. It searches each ancestor whole, so thoughts appended to the parent after the fork point appear in the branch's results too.

Merging is a server operation. It re-appends the source chain's thoughts, unsigned, onto the target and then deregisters the source. It copies type, role, content, importance, confidence, tags and concepts, and maps agents by id. It drops relations, refs, sessions, entity types, source episodes, signatures and original timestamps. It refuses targets that are not open-write chains, and sources that have private segments, governance records or signed thoughts, because merging them would publish private text and discard signatures. A merge therefore preserves what was remembered but not how it was connected or who signed it.

4. Integrity and Access Control

4.1 The Hash Chain

Each thought's `hash` is the lowercase hex SHA-256 of a canonical encoding of the record (`compute_thought_hash`). The canonical form is a fixed struct serialized with bincode 2 in its standard configuration. It contains every field except `hash` itself and two fields excluded for backward compatibility: `entity_type` and `source_episode`. It includes `prev_hash` and the signature fields. Every thought therefore satisfies:

```
thought.hash      == sha256(bincode(canonical(thought)))
thought.prev_hash == previous_thought.hash      ("" for index 0)
thought.index     == its position
```

Every open verifies the whole chain (`verify_integrity`): it checks each index, each `prev_hash` link, and recomputes each hash. This costs O(n) and runs before the chain serves reads. Checking links alone would miss an edited record, which is why the hashes are recomputed.

This is tamper *evidence*. An attacker who can write the file can rewrite it entirely and recompute every hash. Evidence becomes useful when the head hash is recorded somewhere the attacker cannot reach, and signatures (Section 4.2) bind individual records to keys the attacker does not hold. For governed chains MentisDB records the head hash itself, in a checkpoint signed by the operator key (Section 4.4), so a later rewrite of the verified prefix locks the chain instead of passing unnoticed.

Two gaps are known. First, `entity_type` and `source_episode` are not covered by the record hash, so they can be edited in the file without breaking the chain (v2 signatures do cover them). Second, migrating a chain from schema V0 to V2 recomputes all hashes without checking the old ones, so tamper evidence does not extend across that migration.

4.2 Signatures

A thought may carry a detached Ed25519 signature (RFC 8032, strict verification) and the id of the signing key. The signature is checked on append against an active, unrevoked key of the thought's agent, on every chain. A bad signature is always refused, whatever the chain's policy.

Two payload versions exist, distinguished by the stored signature itself:

Stored `thought_signature`	Version	Payload
64 bytes	v1	Legacy JSON of the semantic fields. Not bound to a chain or position, so replayable; accepted only on open chains.
65 bytes starting `0x02`	v2	Domain `mentisdb/thought/v2`, then chain key, agent, `prev_hash`, session, type, role, content, confidence, importance, tags (which include the scope), concepts, refs, relations with validity, entity type and source episode.

The v2 payload (`encode_thought_v2` in `src/signing.rs`) uses big-endian length-prefixed fields and a NUL-terminated domain tag, so a signature for one payload type can never verify as another. Binding the chain key and the previous head means a signed thought is valid at exactly one position of one chain. The cost is a race: if another writer moves the head first, the append fails with the typed error `StaleHead` and the client signs again against the new head. The `id`, `index` and `timestamp` are assigned by the chain and covered by the record hash rather than the signature.

Signing happens on the client. The CLI signs locally with `--private-key-file` and never sends the seed.

4.3 Write Policies Recorded in the Chain

Each chain has a write policy (`AccessMode` in `src/access.rs`):

Mode	A new thought must
`Open` (default)	nothing more; unsigned writes are accepted
`Signed`	carry a valid v2 signature by an active key registered to its agent
`Allowlist`	carry a valid v2 signature by a key in the policy's allowed set that is still active in the registry

The policy is not configuration. It lives in the chain as signed `AccessPolicy` thoughts, replayed at open. Each record holds the mode, the governor key set, an RFC 6962 Merkle root over that set, a sequence number and the hash of the previous policy record. Every change must be signed by a current governor. An invalid record found on replay is ignored for enforcement and reported by `mentisdb chain access verify`.

Access is administered by the *operator*: whoever can reach the dashboard behind its PIN, or run the local `mentisdb` command line with `--offline` against the data directory. Switching a chain between public and private, allowing and denying keys, registering and revoking agent keys and disabling agents happen only there. The REST routes for these answer 403 to every token, and the MCP tools for them are not offered. The operator signs governance records with a key kept in `<data dir>/operator/operator.key` (mode `0600`), which is a governor of every policy the operator writes. That key is the recovery path when every agent governor key is lost, and a record that would leave no governor is refused, so a policy cannot lock itself out. Filesystem access is the trust boundary: whoever has it already holds every byte of the chain.

`Signed` chains trust the agent registry's key list, which is a sidecar outside the hash chain. `Allowlist` chains do not, because their allowed keys are committed in the chain. The full design, formats and threat model are in `docs/CHAIN_ACCESS.md`.

4.4 Public and Private Segments

Policies also decide who may read. A chain is *public* while its policy is `Open` and *private* while it is `Signed` or `Allowlist`. The mode is not a flag but the history of the governance records: every valid policy record is a mode-transition entry that opens a new *segment*, lasting until the next valid record (`src/audience.rs`). Each thought belongs to exactly one segment, decided by its position, and policy records themselves are always private.

```
index:    0 ......... 9 | 10 ......... 49 | 50 ......
record:   (none)        | seq 0 allowlist | seq 1 open
segment:  public        | private         | public
```

The segment map is recomputed from the chain at every open, so nothing is stored separately and older chains with policies gain segments simply by being read. A chain without policy records is one public segment.

There are two audiences. An *authorized* reader sees every segment. A *public* reader sees only public segments, and sees them as if the private thoughts did not exist: BM25 corpus statistics, the recency anchor, invalidation, successor redirection and standing context are computed over the visible set only, and graph expansion never passes through a hidden thought. This holds on every read surface: ranked, lexical, vector and graph search, recent and standing context, traversal, single-thought reads, context bundles, federated search, branch ancestors, Markdown export, listings, counts and the policy and verification reports.

Writes respect the boundary too. An unsigned writer cannot link to, supersede or deduplicate against a thought in a private segment, and a refused write does not reveal which agents or keys exist. Push channels respect it as well. Webhooks never receive private thoughts or governance records, and the MCP event stream carries only `tool_called` events (tool, chain, outcome and duration), never arguments or results.

Some metadata stays public by design, so that the chain remains verifiable. A public reader can see that a chain has private segments and where they are, how many thoughts each gap holds (indices skip hidden positions), and the one `prev_hash` that links the first public thought after a gap to the hidden thought before it. Timing is not padded.

Governed chains are also verified when they open, before anything is served. Every thought in a private segment must carry a valid v2 signature by a key allowed at its position. A thought that fails is quarantined: hidden from every reader and listed by `MentisDb::verification`, `mentisdb chain access verify` and the signature report. To keep later opens cheap, the operator key signs a checkpoint (`<stem>.verified.json`) bound to the chain key, the head hash at that point, the policy's key root and sequence number, and the quarantine list; later opens verify only the thoughts after it. If a valid checkpoint exists but the chain file no longer matches it, because it was truncated or its prefix rewritten, the chain is locked and nothing is served, rather than falling back to looking public. The operator inspects the file and deletes the checkpoint to re-seal the chain.

The library defaults to the authorized audience, because an embedder that holds the chain directory already holds every byte of it. The daemon starts every request from the public audience and raises it only for a proven reader (`src/server_lens.rs`), so a code path that forgets to authenticate loses data instead of leaking it. Private segments are not encrypted at rest. Anyone who can read the files can read them.

4.5 Read Authentication

A reader proves possession of an authorized key with one header per HTTP request, signed locally by the client process (the CLI, the stdio MCP proxy or an SDK), so the private key never travels and never enters a model's context (`src/read_auth.rs`):

```
X-Mentis-Auth: v1 agent=<id>;key=<key id>;pk=<64 hex>;ts=<unix secs>;nonce=<hex>;sig=<128
    hex>

signed bytes: "mentisdb/read-auth/v1\0" METHOD path_and_query sha256(body)
              agent_id key_id public_key ts nonce
```

The signature binds one method, one URL and one body, and therefore one chain key. The server accepts it when the timestamp is within 60 seconds of its clock and the nonce has not been seen in that window, so a captured request can be neither replayed nor turned into a different one. Invalid proofs are rate limited to 30 per minute per peer. The proof only identifies the reader. Whether that key may read a chain is decided per chain (`MentisDb::is_read_authorized`): the key must be active in the chain's registry with exactly that public key, and be in the policy's key set; in `Signed` mode every active registered key qualifies, because each may already write. The proof authenticates but does not encrypt, so private chains should be served over TLS.

Every returned thought carries a `provenance` object: its segment, whether it is signed, whether the signature verifies now against the chain's registry and policy, the verified agent when the signing key is still active, and `trust: "untrusted_data"`. Stored memories are data to weigh, never instructions to follow, and the field says so to the agent reading them.

4.6 Transport Authentication

The daemon binds to `127.0.0.1` by default. Bearer tokens are optional and off by default (`src/auth.rs`). When enabled, tokens are stored as SHA-256 hashes in a `0600` file and compared in constant time. A token is either global or scoped to one or more chains. REST and MCP share one table that marks each operation as chain-scoped or global-only, normalize chain keys before authorization, and authorize the same fields the handler executes. Chain-scoped tools that name no chain require access to the default chain.

While tokens are off, requests that come from a browser (they carry `Origin` or `Sec-Fetch-Site`) must name an allowed host, which blocks DNS-rebinding and cross-site requests to the loopback listeners. The dashboard sits behind an optional PIN with a per-IP limit of five failures per five minutes, sends a strict Content-Security-Policy and frame-denial headers, and guards state-changing calls against cross-site requests.

Several authorization bypasses have been found and fixed, in 0.10.9 and again in the unreleased changes, including decoy chain-key fields that let a chain-scoped token read another chain and a fail-open default for unlisted MCP tools. Deployments that expose the daemon beyond loopback should run current master and enable tokens.

5. Storage and Durability

5.1 Record Format

Each chain is one append-only file, `<stem>.tcbin`. Each record is an 8-byte little-endian length followed by the bincode 2 encoding of the `Thought` (`persist_binary_thought`). A record payload may not exceed 10 MiB.

The file stem is the chain key with characters outside `[A-Za-z0-9_-]` replaced by `_`, a dash, and the first 16 hex characters of the SHA-256 of the chain key. Two keys that sanitize to the same

string therefore still get different files.

A legacy JSON-lines format (`.jsonl`) can still be read for migration but no new chain uses it.

5.2 File Layout

```
$MENTISDB_DIR/                          (default ~/.cloudllm/mentisdb)
  mentisdb-registry.json                chain registry
  mentisdb-skills.bin                   skill registry
  mentisdb-webhooks.json                webhook registrations
  bearer-tokens.json                    hashed bearer tokens (0600)
  tls/cert.pem, tls/key.pem             self-signed certificate
  .fastembed_cache/                     local embedding model
  <stem>.tcbin                          the hash chain (canonical)
  <stem>.agents.json                    agent registry
  <stem>.entity-types.json              entity-type registry
  <stem>.vectors.<model>.<ver>.<dim>d.json       vector sidecar snapshot
  <stem>.vectors.<model>.<ver>.<dim>d.json.wal   its write-ahead log
  <stem>.vectors.<model>.<ver>.<dim>d.hnsw.bin   persisted HNSW graph
  <stem>.vectors.managed.json           which sidecars stay in sync on append
  <stem>.auto_edges.bin                 implicit-edge overlay
  <stem>.verified.json                  operator-signed verification checkpoint
  operator/operator.key                 operator signing key (0600)
  migrations/vX_to_vY/                  archived pre-migration files
```

5.3 Write Path and Durability Modes

Appends go through a dedicated writer thread per chain. Two modes exist:

Mode	Behaviour	What a crash can lose
Strict (`auto_flush = true`, the default)	Each append returns after its group commit is written to the operating system. When other appends are already queued, the writer waits up to 2 ms (`MENTISDB_GROUP_COMMIT_MS`) to batch them.	Nothing on a process crash. On power loss, writes since the last sync unless `MENTISDB_FSYNC=1`.
Buffered (`auto_flush = false`)	Appends are queued (capacity 128) and written every 16 records.	Queued and unwritten records.

`MENTISDB_FSYNC=1` adds one `fsync` per group commit, and registry files are then synced before their atomic rename. Explicit flush, shutdown and drop always sync. A failed group commit is rolled back to the pre-commit file length, so a full disk cannot leave a torn record followed by its duplicate.

Embedding stays off the critical section. A daemon append takes an embedding ticket under the chain's write lock, computes the embedding with no lock held, and then commits the vector under a brief lock. Vectors are applied in chain order even when concurrent appends finish out of order, and a failed embedding schedules the background rebuild instead of blocking the sidecar. With a 5 ms embedding model this cut the lock hold per append from about 9.5 ms to about 0.5 ms (`storage_paths / append_lock_hold`).

The agent registry is not rewritten on every append. Its counters are recomputed from the chain at open, so only identity changes (a new agent, a rename, a learned owner) are written immediately, and counters are saved every 16 appends and at shutdown.

One writer process per data directory is assumed. There is no cross-process file lock.

5.4 Crash Recovery

Opening a chain repairs only what is safe to repair.

- **Torn tail.** If the final record is strictly partial (a short length prefix, or a payload that ends at end of file), it is copied to `<file>.torn-<millis>` with mode `0600` and truncated before the writer reopens the file.
- **Mid-file corruption.** If the bytes after a bad length prefix still decode as a complete record with a 64-digit hex hash, the damage is not a torn tail. The open fails with `InvalidData` and the file is left untouched, so no acknowledged thought is ever discarded as "tail".
- **Bounded decoding.** Every decode of untrusted bytes runs under a budget: 64 MiB per thought record and 1 GiB per derived file (`src/decode_limits.rs`). A forged length prefix yields `InvalidData` instead of an allocation that aborts the process.

These rules are pinned by adversarial tests that cut the last record at every byte (`tests/core_crash_recovery_tests.rs`).

5.5 Derived State

Derived files make search fast, and each one can be rebuilt from the log.

- **Vector sidecars** store one embedding per thought for one model, version and dimension. Changing any of these selects a different file, so incompatible embeddings never mix. Each append writes one integrity-chained record to the sidecar's write-ahead log (magic `MDBVWAL1`). The snapshot is compacted when the log reaches one sixteenth of the entries, and at least 32. A sidecar is fresh only if its thought count, head hash and entry count all match the chain.
- **Background rebuild.** Opening a chain never embeds its history on the request path. A missing or stale sidecar is queued to one process-wide worker, one job per sidecar at a time, built in batches of 32 with one shared model, and adopted when done. Until then ranked search serves lexical and graph signals. Chains of 32 thoughts or fewer embed inline.
- **Implicit-edge overlay** (Section 6.6) records the thought count and head hash it covers. On load it replays a missing tail of up to 2048 thoughts, and otherwise rebuilds.
- **HNSW graphs** start with a magic and a SHA-256 digest of the embedding space and its documents. The digest is checked against the sidecar and the graph body on load; any mismatch, torn write or legacy file triggers a background rebuild.

5.6 Schema Versions and Migration

Version	Change
V0	Original layout, no version field (JSON lines)
V1	Explicit version, registry-backed chains, optional signatures
V2	`Reframe` type, `Supersedes` relation, cross-chain `chain_key` on relations
V3	`valid_at` and `invalid_at` on relations; later `entity_type` and `source_episode` were appended without a version bump

The loader tries the current layout first and falls back record by record to the legacy layouts, so a file with mixed versions still loads. When the daemon starts it migrates registered chains before serving: it writes the new file beside the old one, archives the old file under `migrations/`, and

renames the new file into place. A chain written with the older JSON-based hash is rehashed with the bincode hash on open. Both steps skip work that is already done, so running them twice is harmless. Both also block: a large chain is unavailable until its migration finishes.

A journaled migration engine (`src/migration/`) replaces this pass but is not yet wired into the daemon. It runs each chain through a durable state machine (planned, backed up, staged, committed, finalized) recorded in `migrations/journal/`, keeps the original file under `migrations/backup/`, and streams records through one upgrade step per schema version, so memory stays flat for any chain size. It is stricter than the current pass. A schema 3 record must hash to exactly what it stored, under either hash algorithm, or the migration is refused, so a migration can never turn a tampered chain into a freshly valid one. Only schemas 0 to 2, whose stored hashes cannot be checked, are rehashed. A killed migration resumes or is discarded on the next start, a failing chain is quarantined untouched while the others proceed, an advisory lock keeps two processes off one chain, and a chain written by a newer release is refused rather than modified. Its manager runs on a small worker pool after startup and answers requests for a chain still migrating with HTTP 503, `Retry-After` and the percentage done.

Because bincode encodes enum variants by position, new variants must be appended at the end of an enum and existing ones never reordered. A chain that contains a variant added later, such as an `AccessPolicy` record, cannot be read by an older MentisDB.

5.7 Backups

A backup is a ZIP file named `mentisdb-YYYY-MM-DD-HH-MM-SS.mentis` holding a manifest with the SHA-256 and size of each file, the registries, chain files and sidecars, skills, webhooks and optionally the TLS material. Restore checks the manifest before writing, bounds sizes and file counts, and rejects symlinks and zip bombs. A backup that extends a governed chain fast-forwards it only after every new record verifies against the policy in force at its position (`verify_governed_suffix`); otherwise it is imported as a separate, renamed chain.

6. Retrieval

Two read paths exist. *Filter search* (`query`) returns thoughts that match indexed fields and a case-insensitive substring, in append order. *Ranked search* (`query_ranked`) scores candidates with several signals and returns them best first. Agents use ranked search almost always. This section describes ranked search with its defaults as of current master.

6.1 Candidates

Ranking starts from a candidate set (`retrieval_candidates`). It removes invalidated thoughts unless `include_invalidated` is set, applies `as_of` (Section 3.6), and applies filters on type, role, agent, tags, concepts, scope, entity type, session and time range. Filters only restrict; they never add score. After scoring, superseded thoughts that would have ranked are replaced by their live successors and contested thoughts are preceded by their contesters (Section 3.6).

6.2 Lexical Signal: BM25 with Document-Frequency Gating

Text is normalized by lowercasing, tokenizing and stemming with the Snowball English (Porter2) stemmer. At query time, a table of 325 irregular forms maps words such as “went” and “saw” to their lemmas (“go”, “see”), which stemming cannot do.

Each thought is indexed in five fields, scored with BM25 (`k1 = 1.2`, `b = 0.75`) and weighted:

Field	Weight	DF cutoff
content	1.0	30%
tags	1.6	30%
concepts	1.4	30%
agent id	1.5	70%
agent registry text (name, owner, description, aliases)	1.1	60%

```
idf(t)        = ln( (N - df + 0.5) / (df + 0.5) + 1 )
score(t, f)   = idf(t) * tf*(k1+1) / (tf + k1*(1 - b + b*len_f/avg_len_f)) * weight_f
```

MentisDB has no stop-word list. Instead, for each query term and each field, the term is skipped in that field when `df / N` exceeds the field's cutoff. This *document-frequency gate* is a hard cliff rather than a down-weight: a term in 29% of thoughts scores fully in `content`, and one in 31% scores nothing there. A gated term can still score through a field whose cutoff it respects. Details that matter in practice:

- By default `df` counts thoughts that contain the term in *any* field (global scope), and that one ratio is compared with each field's cutoff. Registry text is attached to every thought of an agent, so a word in an agent's description counts as present in all of that agent's thoughts. On one real 750-thought chain, project words that occur in 2% to 9% of the content were gated out of it by registry text alone.
- A per-field scope that counts each field separately exists as an opt-in (`MENTISDB_DF_SCOPE=field`, or `df_scope` per query). It is not the default because it brought no gain: under the current defaults both scopes score the same on LoCoMo and LongMemEval, and on a real chain it moved citation-style probes by two points in opposite directions at rank 1 and rank 10.
- Gating is off for chains under 20 thoughts. `N` counts every committed thought, including invalidated ones and ones excluded by filters. The IDF of a surviving term always uses the global `df`.

Registry text is part of the index, so when an agent's description, owner, display name or aliases change, the lexical index is rebuilt and older thoughts become findable by the new text.

Synonyms. Query terms are expanded with a built-in thesaurus of 895 headwords, keyed by stem so inflected forms expand too. Each term gets at most three synonyms at weight 0.3, in the library and the daemon alike. On the A/B chains of Section 9.2, turning synonyms off scored 75.3% LoCoMo R@10 and 69.6% LongMemEval R@5, weight 0.3 scored 75.2% and 70.0%, and the older weight 0.7 scored 74.8% and 68.2%, so 0.3 is kept mainly for LongMemEval.

6.3 Vector Signal and Fusion

The default build includes local embeddings through fastembed: the `all-MiniLM-L6-v2` model, 384 dimensions, run on ONNX on the CPU. A dependency-free `local-text` provider (256 dimensions) is the fallback. `MENTISDB_EMBEDDINGS` selects `auto` (default), `fastembed`, `local-text` or `off`.

The query is embedded and compared by cosine with each thought's vector. Up to 256 hits per sidecar are kept, and hits under cosine 0.04 are dropped. The vector score is `0.5 * cosine`. It is fused with the lexical score by:

```
rel                = lexical / best_lexical_score_of_this_query     (0 to 1)
vector_contribution = vector * (1 + 35 * exp(-rel / 0.8))
```

With no lexical match the vector signal is amplified 36 times, so a paraphrase with no shared words can still rank. At the query's best lexical match it is amplified about 11 times. Because the decay depends on the lexical score *relative to the best one*, it does not depend on query length.

That is the point of the design. The earlier rule, `1 + 35 * exp(-lexical / 3)`, decayed on the raw BM25 score, which grows with the number of query terms. A ten-word paraphrase gave even weak lexical matches a score of ten or more and silenced the vector channel exactly when the wording differed, which the real-chain audit measured (Section 9.4). The old rule remains available as `MENTISDB_VECTOR_FUSION=absolute`. The decay rate 0.8 was chosen from a sweep (Section 9.2): larger rates gain a little more recall but order results almost purely by vectors and lose lexical precision.

Below 50,000 vectors the search is an exact scan. At or above that threshold (`MENTISDB_HNSW_THRESHOLD`) it switches to an HNSW graph (M 48, ef_construction 400, ef_search 128). A filtered HNSW search scans exactly when the allowed set is small (at most 4096 thoughts or under 5% of the corpus) and otherwise oversamples, so filters do not lose matches the graph skipped.

6.4 Graph Expansion

Graph expansion is off unless the query sets `graph`. When on, the top 20 lexical hits seed a breadth-first traversal over explicit relations and the implicit-edge overlay, bidirectional by default, to depth 2, visiting at most 128 thoughts (`expand_with_overlay_classified`). Each reached thought gets three scores:

- `graph = 1 / depth`,
- `seed_support = 0.2 * min(paths - 1, 5)` for thoughts reached from several seeds,
- `relation = boost(kind) / depth` for the strongest edge kind on its path.

Relation kind	Boost
ContinuesFrom	0.60
BranchesFrom	0.55
Corrects, Invalidates	0.50
Supersedes	0.45
DerivedFrom	0.40
Summarizes, CausedBy	0.20
Supports, Contradicts	0.15
RelatedTo (implicit)	0.08
References	0.06

6.5 Session Cohesion, Importance, Confidence and Recency

Session cohesion surfaces the turns around a good match, which often hold the answer without sharing its words. Every candidate with a lexical score of at least 3.0 is a seed. A candidate whose own lexical score is under 5.0 and that lies within 12 positions of a seed receives `1.2 * (1 - d / 12)`, where `d` is the distance in append order. This is always on (`apply_session_cohesion`).

Importance shifts close races towards what the author marked as important: `lexical * ( importance - 0.5) * 0.3` when there is a lexical match, else `(importance - 0.5) * 0.1`. **Confidence** adds `0.1 * confidence`. **Recency** adds at most 0.05, linear in append position; there is no time-based decay. Recency is deliberately a tie-breaker so that old memories are not pushed down.

The total is:

```
total = lexical + vector_contribution + graph + relation + seed_support
      + importance + confidence + recency + session_cohesion (+ rrf)
```

6.6 Implicit Edges

Most agents write few explicit relations, which leaves graph expansion little to walk. The implicit-edge overlay adds `RelatedTo` edges between thoughts whose embeddings have cosine at least 0.85, keeping the 5 strongest per thought (`MENTISDB_AUTO_EDGE_THRESHOLD` , `MENTISDB_AUTO_EDGE_K`). Up to 128 thoughts the build compares all pairs. Above that it runs one nearest-neighbour search per thought through the vector backend. Each append adds the new thought's edges with one linear scan. Chains above 50,000 thoughts get no overlay. The overlay is saved every 32 appends and is never part of the hash chain.

6.7 Optional Reciprocal Rank Fusion

With `enable_reranking` , the top 50 candidates are ranked three times: by lexical score, by vector score, and by graph-related score. Only candidates with a positive score enter each list. Each candidate then gains a consensus bonus:

```
rrf = 50 * sum over lists containing it of 1 / (60 + rank)
```

A candidate ranked first in all three lists gains about 2.5. RRF is off by default. Before 0.10.9 it replaced the additive total and collapsed rankings; it now only adds.

6.8 Explaining a Ranking

Each hit carries its full score breakdown: `lexical` , `vector` , `graph` , `relation` , `seed_support` , `importance` , `confidence` , `recency` , `session_cohesion` , `rrf` and `total` , plus the matched terms, the fields they matched in, and for graph hits the distance, seed path count, relation kinds and path. A human or an agent can see why any thought ranked where it did.

Context bundles (`query_context_bundles`) group results by seed: each lexical seed comes with the thoughts graph expansion reached from it and the path that reached them, so an agent can read supporting memories in provenance order.

6.9 Deduplication on Write and Collapse at Read

Automatic deduplication is off unless `MENTISDB_DEDUP_THRESHOLD` is set; 0.85 is a suggested value, not a default. When on, each unsigned append compares the new thought's normalized token set with the last 64 thoughts (`MENTISDB_DEDUP_SCAN_WINDOW`) by Jaccard similarity. If the best match reaches the threshold, the new thought gets a `Supersedes` edge to it, which hides the older one by default. A prior thought the author rated more important than the new one is never superseded, so a throwaway restatement cannot hide a critical constraint. Signed thoughts are never deduplicated.

The window is a real limit: a fact re-recorded hundreds of thoughts later is not caught. Section 9.4 shows that duplicates are a measured problem on real chains.

Ranked search therefore also collapses duplicates when it builds a result page. For text queries, a hit whose normalized token set has Jaccard similarity of 0.9 or more with a higher-ranked hit is dropped from the page, and its id is listed on the survivor (`collapsed_duplicates`). Nothing is written to the chain. Collapse is on by default and can be turned off with `MENTISDB_COLLAPSE_DUPLICATES =off` or per request. It costs about 0.3 ms per query on a 5000-thought chain.

6.10 Standing Context and Recent Context

Memory has two halves. The episodic half is what happened, in order. The semantic half is what stays true. Recent context (`mentisdb_recent_context`, 12 thoughts by default) serves the first. It forgets: a preference from three months ago has scrolled out of any practical window.

Standing context (`MentisDb::standing_context`, `src/standing.rs`) serves the second. It selects live thoughts of the standing types, ranks them by importance and then recency, collapses restatements, and caps the result:

Group	Types	Cap
Constraint, Preference, User trait, Relationship, Goal, Decision, Lesson	`Constraint`, `PreferenceUpdate`, `UserTrait`, `RelationshipUpdate`, `Goal`, `Decision`, `LessonLearned`	8 per type, 40 in total
Handoff	`Handoff`	the newest one
Checkpoint	`Checkpoint` or `Summary` type or role	the newest one

Only user-scoped or unscoped thoughts are standing. Within a group, restatements with Jaccard similarity of 0.85 or more collapse to the newest. A superseded constraint or decision is replaced by its live successor, even when the successor is not itself a standing type such as a `Correction`, and contested entries are flagged in the prompt. The result is a compact prompt block delimited by `=== STANDING CONTEXT ===` that the agent loads at startup, next to the recent-context block. It is available as an MCP tool, a REST route (`POST /v1/standing-context`) and a CLI command (`mentisdb thought standing`).

6.11 Tuning Per Query

Every retrieval default in this section can be overridden per query, on REST, MCP and the CLI as well as in the library: successor redirection, duplicate collapse, vector fusion, synonym weight, document-frequency scope and the per-field cutoffs. Environment variables set the process-wide defaults.

6.12 What Is Not in the Default Path

The crate contains components that ranked search does not use by default, and this paper makes no claim based on them: personalized PageRank (`src/search/ppr.rs`), pseudo-relevance feedback (`query_expansion.rs`), a query-intent classifier (`query_intent.rs`) and embedding synonyms. Personalized PageRank and the LLM-reranking fields are deprecated no-ops. LLM memory extraction (`src/llm.rs`) exists as an explicit, opt-in operation that writes `LLMExtracted` thoughts.

7. The Agent Protocol

A memory store is only as good as the habits of the agents that use it. MentisDB ships its operating protocol as a skill (`MENTISDB_SKILL.md`), served to MCP clients as the resource `mentisdb://skill/core` and pointed to by the server's `initialize` instructions. Its main rules are these.

Startup. List chains and pick the project's chain, bootstrap it (which writes a genesis checkpoint only on an empty chain), load the skill, list agents and reuse an existing identity, load standing context, load recent context, then write a checkpoint. Agents never invent a new agent id; they ask the user.

Search before write. Before every append the agent runs a ranked search for its proposed memory and makes three decisions. Should it write at all, or is this already known? Should it

correct, invalidate or supersede an earlier thought instead of writing a parallel one? Which one to three earlier thoughts should it link to, with which typed edge? This keeps chains small and connected, and it is the main defence against duplicates. Automatic deduplication is only the downstream half. A change of policy or decision must carry a `Supersedes`, `Corrects` or `Invalidates` relation to the thought it replaces. A plain reference only marks the old version contested, and if the append response carries warnings the agent adds the typed link.

Write triggers. Write immediately on a lesson learned, a decision, a constraint, a correction or invalidated assumption, a checkpoint, a completed task, an open question or a hypothesis. One strong memory is better than many weak ones.

Recall triggers. Search before answering when a question touches the user's preferences, a past decision, a constraint or how something was done before, or when the user says "again", "remember" or "last time". The agent checks each hit's `version_status` and answers from the current version when a hit is contested or superseded. Searching with `include_invalidated=true` is for history, not for finding the current answer. Never say "I don't know your preference" without having searched.

Standing context covers what must be known at the start. Recall triggers cover what becomes relevant later. Together they address the complaint that users have to repeat themselves.

8. Interfaces

MCP. The daemon serves the Model Context Protocol over streamable HTTP. A stdio mode lets clients that only speak stdio connect through a local proxy that starts the daemon if needed. The server exposes 44 tools covering bootstrap, append and retrospectives, filter and ranked search, standing and recent context, context bundles, traversal, federated search across chains, branches and merges, agent registry reads and descriptions, access policy reads and signature verification, entity types, import and export, vectors, webhooks and the skill registry. Key and policy administration is not among them.

REST. A versioned API under `/v1/` mirrors the tools with 51 routes and can be embedded in another server with `rest_router`.

CLI. The `mentisdb` binary is also a client for every MCP tool, REST route and dashboard action, with typed argument validation and `--help` examples on each command (`docs/CLI_PARITY.md` maps each one and a test keeps the map in sync). For example:

```
mentisdb thought append "Use Postgres for the billing service" \
  --chain myproj --agent dev --type Decision
mentisdb search "which database did we pick for billing" --chain myproj
mentisdb thought standing --chain myproj
mentisdb chain access verify --chain myproj
```

Dashboard. A local HTTPS web interface browses chains, thoughts, agents, skills and access policies, manages settings, and is one of the two operator surfaces for access administration. Without a PIN it neither administers access nor shows chains with private segments.

Skill registry. Agent instruction bundles are versioned in `mentisdb-skills.bin`. The first version is stored whole and later versions as diffs. A skill can be deprecated or revoked with its history kept, or deleted permanently. An agent with registered keys must sign its uploads. The registry is integrity-checked as a file; it is not a hash chain.

Webhooks and setup. Webhooks notify external services of appends. `mentisdb setup` configures MCP clients: Claude Code, Claude Desktop, Codex, Copilot CLI, VS Code Copilot, Gemini, Grok, OpenCode and Qwen.

9. Evaluation

9.1 Setup

Two public benchmarks measure retrieval recall: for each question, does the gold evidence turn appear in the top k thoughts? Both harnesses are in the repository (`locomo-benches/`, `lme-benches/`) and run against a live daemon with a fresh chain.

- **LoCoMo-10P**: ten long multi-session conversations, 1977 questions, split into single-hop (1554) and multi-hop (423). Turns are ingested with `ContinuesFrom` edges between consecutive turns, speaker and turn tags, and importance 0.8 or 0.2 by speaker. Queries use ranked search with graph expansion to depth 3. A hit is substring containment of the gold evidence in a returned thought.
- **LongMemEval**: 500 questions over a chain of about 10,900 turns, scored at R@5, R@10 and R@20.

All runs use the daemon's automatic thesaurus. The embedding model differs between runs and is stated with each result.

9.2 Results

The full protocol was re-run on master on 2026-11-01 with the current defaults (relative fusion at 0.8, duplicate collapse at 0.9, synonym weight 0.3, global document frequency), MiniLM embeddings, a fresh daemon and data directory, and fresh ingestion. The scripts and question sets are the same as for the release run below. Raw outputs and the protocol are in `results/benchmarks/2026-11-01-full-protocol/`.

Benchmark	Pipeline	R@5	R@10	R@20	Queries/s
LoCoMo-10P	default	67.9%	**76.7%**	83.4%	428
LoCoMo-10P	with RRF	68.6%	76.9%	83.5%	429
LoCoMo-10P single-hop	default	71.9%	80.4%	86.0%	
LoCoMo-10P multi-hop	default	53.0%	62.9%	73.8%	
LongMemEval	default	**69.8%**	75.6%	81.4%	371
LongMemEval	with RRF	70.2%	75.8%	82.0%	362

LongMemEval R@10 and R@20 are the per-type results weighted by the number of questions of each type. LoCoMo R@5 comes from a separate top-5 evaluation of the same chain.

For comparison, release 0.10.9.54 (October 2026) under the same protocol, before valid-version retrieval, relative fusion and collapse:

Benchmark	Pipeline	R@5	R@10	R@20
LoCoMo-10P	default	63.9%	73.5%	
LoCoMo-10P	with RRF	66.0%	74.1%	
LongMemEval	default	68.0%	72.8%	78.8%
LongMemEval	with RRF	68.6%		

The default pipeline gained 4.0 points of LoCoMo R@5, 3.2 of LoCoMo R@10, 1.8 of LongMemEval R@5 and 2.8 of LongMemEval R@10. RRF now adds less than a point, so it stays off by default. Multi-hop questions remain about 17 points behind single-hop ones.

Choosing the fusion decay. The decay rate of relative fusion was chosen by an A/B on scratch copies of older, already-ingested benchmark chains (the LoCoMo one with 256- dimension `local-text` embeddings), comparing configurations on one binary. Those numbers measure the effect of each setting, not an absolute result, and are not comparable with the tables above:

Configuration	LoCoMo-10P R@10	LongMemEval R@5
Absolute fusion (previous default)	73.1%	68.0%
Relative fusion, decay 0.10	73.0%	67.8%
Relative fusion, decay 0.25	73.6%	68.4%
Relative fusion, decay 0.40	74.0%	69.0%
Relative fusion, decay 0.80	75.2%	69.4%
Relative fusion, decay 1.5	75.5%	70.2%
Relative fusion, decay 3.0	75.9%	69.6%
Relative 0.8 with duplicate collapse	75.2%	70.0%

Larger decay rates gain a little recall but order results almost purely by vectors, so 0.8 was kept for lexical precision. Duplicate collapse alone moved LongMemEval R@5 from 68.0% to 68.4% and left LoCoMo unchanged, because the benchmark conversations contain few repeated turns. Its main effect is on real chains (Section 9.4).

Earlier full runs show where the gains came from:

Release	Main change	LoCoMo-10P R@10	LongMemEval R@5
0.8.0	baseline		57.2%
0.8.0	Porter stemming		61.6%
0.8.1	smooth fusion, cohesion, DF gate	74.2%	67.6%
0.8.9	irregular lemmas	71.9%	66.8%
0.9.9	automatic thesaurus, real embeddings	72.6%	66.8%
0.10.9.54	stemmed thesaurus keys, damped synonyms, RRF fix	73.5%	68.0%
master, 2026-11-01	valid-version retrieval, relative fusion, collapse	76.7%	69.8%

On a two-conversation LoCoMo subset, 0.8.1 reached 88.7% R@10. That number is for the subset only and should not be compared with full-benchmark results.

9.3 Where Misses Come From

A near-miss analysis of 0.8.5 on LoCoMo-10P bucketed the misses at R@10. A quarter of them (25.8%) were found by R@20, which is a ranking error. Of the rest, 43.3% were not found even by R@50, because the evidence shares almost no words with the question. A similar analysis of 0.9.8 on LongMemEval found that 65.7% of misses were not in the top 20. Multi-hop questions remain the weakest category on LoCoMo, about 17 to 20 points below single-hop. The lesson is that lexical retrieval has a hard ceiling on paraphrase, and a small embedding model only partly lifts it.

9.4 Recall on Real Project Chains

Benchmarks use conversations, not the memories coding agents actually write. The `examples/recall_audit.rs` tool audits a copy of a real chain. It builds synthetic queries from the chain's

own thoughts, runs them through the production ranked search with the daemon's defaults, and reports where each target lands, bucketed by its age. Query kinds range from easy (`drop50`: the target's words with half removed) to hard (`ref`: the opening of a later thought that refers to the target, and `sem`: the opening of the target's nearest semantic neighbour among thoughts sharing few of its words).

The audit ran on 2026-10-31 against copies of three real project chains of one user, with 750, 525 and 86 thoughts written by 23, 23 and 5 agents over five to ten weeks. It used release 0.10.10.55 scoring, unchanged, with the daemon's thesaurus expansion, 300 targets per query kind, seed 7 and queries of at most 32 words.

Probe kind	750 thoughts	525 thoughts	86 thoughts
`drop50`: half the target's words	76 / 100	95 / 99	87 / 100
`rare`: the six rarest words	77 / 100	94 / 99	90 / 100
`ref`: opening of a later thought citing it	19 / 65	20 / 49	100 / 100
`sem`: same topic, few shared words	6 / 43	9 / 55	24 / 84
`tail`: words beyond the 512-token window	76 / 100	80 / 93	none

The cells are hybrid R@1 / R@10 in percent. Lexical-only ranking was within one point everywhere except the 15 tail probes of the 525-thought chain, so vectors added almost nothing at these query lengths. The findings:

1. **No age effect on recall at ten.** On the two large chains, `drop50` and `rare` probes reached R@10 of 98% to 100% in every age decile, and `sem` recall was the same for the older and newer halves (46% against 40%, and 56% against 54%). Rank 1 did depend on age: on the 750-thought chain, the older half reached R@1 of 56% against 94% for the newer half. That gap is a duplicate effect, described below, not forgetting. Ranking does not bury the past; loading only recent context at startup did, which standing context addresses. The 86-thought chain showed a real `sem` gap between halves (74% against 97%), but on fewer than 40 probes per half.
2. **Paraphrase is the weak spot.** On `sem` probes the hybrid ranker recalled the target in the top ten for 43% of queries on the 750-thought chain and 55% on the 525-thought chain. Vector similarity alone did better, at 57% and 59%. The reason was the absolute fusion rule then in force (Section 6.3): a long query that shares a few words with many thoughts earns a moderate lexical score, and `exp(-lexical / 3)` then removes almost all of the vector signal. With 8-word queries the hybrid ranker matched or beat vectors alone.
3. **Duplicates and dominant thoughts crowd results.** None of the chains had automatic deduplication on, and only one `Supersedes` edge existed among them. On the 750-thought chain, 301 of 749 live thoughts sat in 145 clusters of near-copies (token Jaccard of 0.85 or more). Most were exact copies re-recorded about 128 positions later, outside the 64-thought dedup window; 149 of the 172 duplicate pairs lay beyond it. The newer copy wins rank 1 by its recency bonus of about 0.009, which explains the rank-1 age gap above. On the 525-thought chain a 13-member cluster restating one build rule, each copy at importance 0.95, took rank 1 in 14 probes meant for other thoughts. On the 750-thought chain single high-importance thoughts took rank 1 in up to 7 probes each. Turning on RRF cut the rank-1 misses caused by a newer near-duplicate from 140 to 9 on that chain, but it only reorders ties and does not change recall at ten.
4. **The document-frequency gate is not what limits recall.** It removed 0.4% to 1.5% of the vocabulary, mostly generic words, and a few domain words such as "android", "commit", "search" and "test". Registry text alone pushed 8 terms over the cliff on the largest chain; one word in 4% of content reached 98% of thoughts that way. Yet per-field

counting and removing the gate both moved R@1 and R@10 by at most about three points, with mixed signs, and no query was left without a scorable term.

5. **Long thoughts are truncated for embedding but still found.** The encoder sees at most 512 tokens and was trained on 256. On the largest chain, 9 of 40 decisions exceeded 512 tokens, losing half of that type's tokens to the vector signal. Queries built from the truncated tails still reached R@10 of 93% to 100%, because the lexical index covers the whole text.
6. **Registry text dilutes embeddings measurably but harmlessly.** Every embedded text ends with the agent's registry description. That raised the mean cosine between unrelated thoughts of the same agent by 0.04 to 0.08, but changed vector-only R@1 by at most three points, not consistently downwards.
7. **One sidecar was stale.** On the smallest chain the newest 9 of 86 thoughts had no embedding and were findable only lexically until the sidecar was rebuilt.

The probes are synthetic and built from each chain's own thoughts, the three chains come from one user, `sem` is a topic-level stand-in for a paraphrase rather than a true paraphrase set, and `ref` probes are noisy. The numbers show where to look, not how every installation behaves.

Two changes followed from the audit: relative vector fusion (Section 6.3) and result-time duplicate collapse (Section 6.9). Re-running the audit on the 750-thought chain with both, and counting a target as found when it was folded into a surviving copy, raised hybrid R@1 from 75% to 98% on `drop50` probes and from 76% to 98% on `rare` probes, and `sem` R@10 from 43% to 52%. Paraphrase remains the weakest case. The remaining limits set the priorities in Section 10.

9.5 Cost

Operation	Measurement
Unsigned append, open chain (buffered, CPU only)	about 55 µs
Signed append (one Ed25519 verification)	about 75 to 80 µs
Append with a 3000-thought managed sidecar	2.1 ms
Rare one-term lexical query, 100,000 documents	1.8 µs
Three-term lexical query, 100,000 documents	0.93 ms
100 vector queries, 10,000 x 128 dimensions, exact scan	33.8 ms (recall@10 100%)
100 vector queries, 10,000 x 128 dimensions, HNSW	8.5 ms (recall@10 91.4%)

The first five figures are from the Criterion benches `access_policy` and `storage_paths` on Apple silicon in release mode, as recorded in the changelog and `docs/CHAIN_ACCESS.md`. The vector figures are from `hnsw_comparison`, re-run for this paper on an Apple M4 Max with synthetic unit vectors. HNSW is about four times faster here; it only becomes the default backend at 50,000 vectors, where an exact scan grows linearly. With `MENTISDB_FSYNC=1` the disk flush dominates append cost.

10. Limitations and Future Work

10.1 Limitations

- **Paraphrase recall.** On real chains, paraphrase-like queries find the target in the top ten only about half the time, even after relative fusion. The default encoder is a small model whose training window is shorter than long thoughts.
- **Duplicates.** Write-time deduplication is off by default and sees only the last 64 thoughts, so restated facts still accumulate on disk. Result-time collapse hides them from search pages, but they still inflate document frequencies and the chain.

- **Global document frequency.** The default gate counts registry text as present in every thought of an agent, which removes some project words from content scoring. The per-field alternative measured no better.
- **Currency depends on links.** Only typed `Supersedes`, `Corrects` and `Invalidates` edges hide an old version. Untyped replacements are detected by a type-based rule and only marked contested, and a replacement with no link at all is caught only by the append-time warning.
- **Integrity gaps.** Two fields are outside the record hash, legacy migration rehashes without checking, and the agent registry is outside the chain, so `Signed` chains trust a sidecar. Integrity is local: there is no consensus, and only governed chains anchor their head hash, in a checkpoint signed by a key on the same host.
- **Read control covers private segments only.** Public segments are readable by anyone who can reach the API, and bearer tokens are off by default. Private segments are not encrypted at rest, a key that can read is also a key that can write, and a public reader still learns where private segments are and how large they are.
- **Blocking migrations.** Until the background migration manager is wired into the daemon, schema and hash migrations still run before a chain serves and rewrite the whole file.
- **Merges lose structure.** A merge drops relations, signatures and timestamps.
- **Single writer.** Two processes writing one data directory are not detected.
- **Benchmarks measure retrieval, not answers.** Recall at k says whether the evidence was retrieved, not whether the agent used it well.

10.2 Future Work

- **Consolidating duplicates on disk:** turn result-time clusters into explicit `Supersedes` edges, under the importance guard, so chains shrink as well as read cleanly.
- **Measuring valid-version retrieval** on benchmarks with changing facts, and on real chains with the correction audit (`examples/correction_audit.rs`).
- **Better paraphrase recall:** evaluate larger local encoders, embed long thoughts in chunks.
- **Per-field document frequency** as the default once benchmarked.
- **Background migrations in the daemon:** wire the migration manager into startup, the CLI and the API, and extend hash coverage to every field.
- **Anchoring head hashes** externally so that a full rewrite of a chain becomes detectable.
- **Answer-level evaluation** with an LLM reader, alongside recall.

11. Related Work

Several systems give agents long-term memory. Mem0 extracts and updates memories with an LLM over a vector store. Graphiti, behind Zep, builds a temporal knowledge graph with LLM extraction and validity intervals. Letta, formerly MemGPT, lets the model page memory in and out of its own context. These systems put an LLM on the write path, which gives them structured facts at the cost of latency, money and a second model's errors.

MentisDB makes the opposite trade. It keeps what the agent wrote, verbatim and typed, adds no model to the write path, and puts the work into retrieval, the write protocol and integrity. Its distinguishing properties are the hash chain, signatures bound to chain position, write policies governed inside the chain, standing context, and fully decomposed scores, all in one embedded binary. Validity intervals on relations are shared with Graphiti's design. Hybrid lexical and vector retrieval is common to most systems.

12. Conclusion

MentisDB treats agent memory as a verifiable log of typed thoughts and builds retrieval, currency and governance on top of it without an external service. Its defaults are stated in this paper exactly, its benchmark numbers come with their setup, and its weaknesses are measured: paraphrase recall and duplicates, not age, are what limit recall on real chains. That is where the next work goes.

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